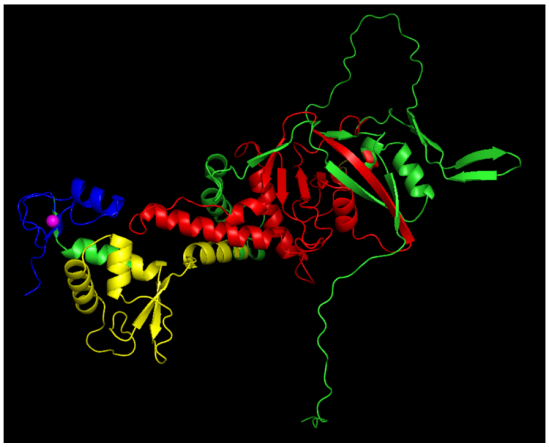
Quick Report: initial findings on aRES



Zn ribbon 1–48 blue, NTD-1 49–120 yellow, RES 126–447 red.

Charge composition, Lys+Arg against Asp+Glu:

Region	Len	K	R	D	E	Net (/100)
Zn ribbon 1–48	48	10	3	2	0	**+11** (+22.9)
NTD-1 49–120	72	0	3	11	8	**−16** (−22.2)
N-block 1–120	120	10	6	13	8	−5 (−4.2)
RES side 126–447	322	23	19	24	21	−3 (−0.9)

The two N-terminal modules carry equal and opposite charge density. The ribbon has ten lysines and no glutamate. NTD-1 has nineteen acidic residues and no lysine. The RES domain is at −0.9 per 100, so the extremity is confined to the N-block.

Residues 31–40 are `CSYC KRKRK N`, a basic run directly after the second CxxC.

Two N-terminal domains

Through InterPro, there seems to be a functionally uncharacterised N-terminal domain (NTD-1) often occurring in front of HEPN and RES domains. The sequence also shows a $(Cys)_4$ pattern consistent with a zinc ribbon: in the AlphaFold model (aRES cofold with Zn) the four cysteines sit in a tetrahedral arrangement at a mean pLDDT of 90.8, and a zinc ion is coordinated at 2.24–2.25 Å.

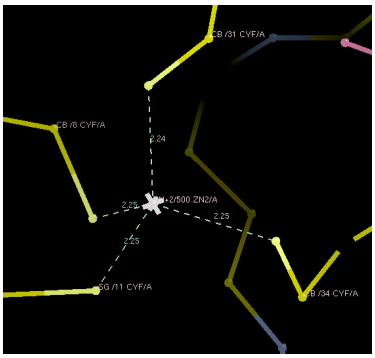
1. NTD-1



Domain	Range	Note
PF18870	49–136	found in other phage defense proteins but is functionally uncharacterised
PF08808 RES	199–363	the NADase

2. Zinc ribbon

Cysteines	**C8, C11, C31, C34** (spacing 3, 20, 3)
Pairwise SG–SG (metal-free)	**3.55–3.81 Å**, mean **3.67 Å**
Zn–S (metal-bound)	**2.24–2.25 Å**
Ideal $Zn(Cys)_4$ edge	**3.76 Å**, deviation **0.09 Å**
Mean pLDDT of the 4 Cys	**90.8**
Not disulfides	an S–S bond is 2.05 Å
Control (T7 gp5)	10 Cys, **no cluster**; closest 7.85 Å



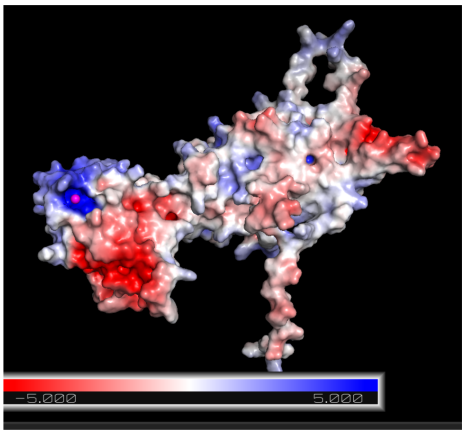
$Zn(Cys)_4$ coordination in the metal-bound model.

Together they form a bipolar surface

The basic ribbon face and the acidic NTD-1 face sit on adjacent lobes and each faces solvent. They do not neutralise one another.

Hypothesis. The N-block recognises a polymerase engaged with nucleic acid:

- the basic zinc ribbon could bind the DNA;
- the acidic NTD-1 could bind the polymerase's basic cleft as a DNA mimic.



Electrostatic surface, $\pm 5\, k_B T/e$.